



GUJARAT POLICE INNOVATION CHALLENGE 2026 · CCTV INTEGRATION HACKATHON

One platform over Gujarat's CCTV — without asking 26 departments to replace theirs

A federated camera registry and GIS, live analytics at the edge, watchlist alerts and cross-camera vehicle tracking, running today on the 30-camera government feed.

THE PROBLEM

Many systems, no shared view, no shared memory

26

government departments, each with its own CCTV and VMS

~80,000

cameras to reach statewide, analog and IP, many vendors

~1,000 km

between the furthest sites, on links that are not uniform

- Vendors, AMC periods and storage differ; retention runs from 7 days to 15 or more.
- No department can see another's cameras, or check what a camera sees against the records that would make it actionable.
- The brief: one searchable platform, analysed and alerting in real time, that grows toward 80,000 cameras without a redesign.

The constraint that shapes everything

Departments keep running their own systems. The first department to join gets value on day one, and the twenty-sixth changes nothing for the first.

A hybrid: Model 1 foundation, Model 3 integration, Model 2 console

1

Registry & GIS

Adopted in full

Mandatory, and correct: nothing else works without knowing which cameras exist, where, and who owns them. Useful on day one with no analytics at all.

3

VMS federation

The integration

One adapter per vendor, normalising to one camera model. Vendor differences stop at the adapter; a department joins without touching anything downstream.

2

Unified viewing

As the surface

One console for operators — but it talks to the platform, never directly to 26 VMS, so a new vendor is never a change to the screen people use.

4

Central VMS

Rejected

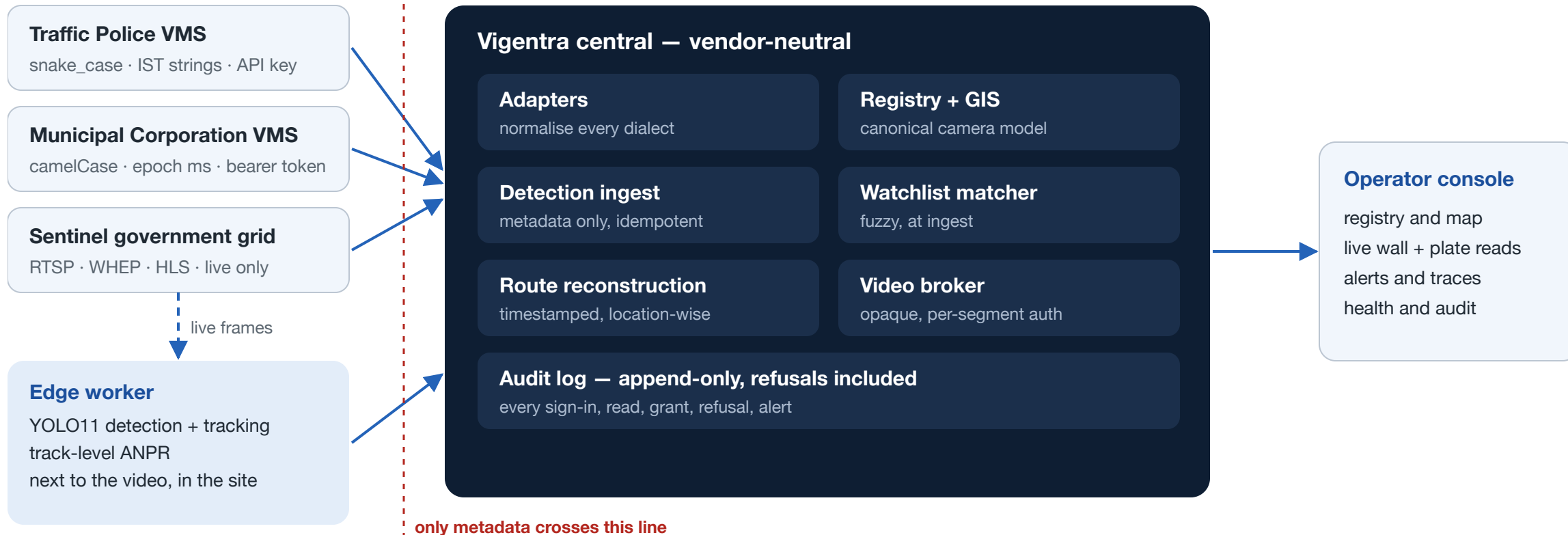
Streaming 80,000 cameras centrally is ~160 Gbps and ~1.7 PB a day, and puts every camera credential in one place. Inference goes to the video instead.

Video is pulled on demand, per authorised session. Analytics run next to the cameras and send only metadata.

SOLUTION OVERVIEW

Frames stay where they are; metadata travels

DEPARTMENT SYSTEMS



One adapter per vendor; above it, every camera looks the same

	TRAFFIC POLICE	MUNICIPAL CORPORATION	SENTINEL GRID
Fields	flat snake_case	nested camelCase	minimal catalogue
Time	IST wall-clock strings	epoch milliseconds	absent — PTS from the stream
Auth	API key header	bearer token	one session per account
Register / archive	yes / yes	yes / yes	none / none — read-only, live-only in code

Analog and legacy

An analog camera is reached through the DVR or encoder that digitises it, which exposes RTSP or ONVIF like any IP source.

Outages stay local

One department down leaves the rest working; its cameras go offline with a stated reason, and every failed poll is audited.

Liveness follows the video

If the grid's web gateway refuses but its video server answers, cameras stay online — this recovered 72 of 180 lost minutes on 14 Sep.

ANPR that reads a vehicle, not a frame

1 · Detect & track

YOLO11 + ByteTrack
give each vehicle a
stable id



2 · Find the plate

only on vehicles ≥ 96
px wide, where a plate
can be read



3 · Bank the crops

every crop stamped
with frame, PTS and
quality



4 · Restore

rectify, register, fuse,
deglare, deblur



5 · Read

CRNN readers on every
variant; a second
reader fills gaps



6 · Vote

Indian plate grammar +
consensus across the
track

Grammar, not guesswork

A character's position says letter or digit, so
GJ05JV34S6 can only be GJ05JV3456.
Impossible states and districts are dropped at
the edge.

One vehicle, one reading

Emitted once per vehicle with the number of
frames that agreed. Unconfirmed readings are
shown as unconfirmed, never as fact.

Object detection

Person, car, motorcycle, bus, truck, auto-
rickshaw, bicycle — with a frame-quality router
that skips frames too degraded to mean
anything.

Skipping plate search on vehicles too small to read gave the same 36 readings on the labelled Delhi clip in 586 s instead of 878 s.

Known at the moment of the read; traced across cameras



Matched at ingest

- Every plate reading is checked against the active watchlist as it arrives — stolen, wanted, missing, blacklisted.
- Fuzzy on purpose: a known OCR confusion ($B \leftrightarrow 8$, $O \leftrightarrow 0$) costs 0.35, an unrelated character 1.0.
- Alerts say whether the match was exact, and near matches are shown, not hidden.



The route, from the plate

- Given a registration: every permitted sighting in time order, with distance, elapsed time and speed per leg, on the map.
- Repeated reads at one camera collapse into one pass; impossible legs are flagged, never dropped.
- Every trace needs a written reason, recorded against the account.

Seconds, not minutes

A plate is uploaded with the frame that settled it; the console refreshes every 5 s.

Prioritised

critical exact stolen or wanted ·
high any other exact · **review** near match. Open alerts lead, most urgent first.

Leaves the screen

Each alert is also POSTed to a webhook, HMAC-signed — dispatch, SMS or messaging sit behind it.

Closed by a person

Acknowledged or dismissed with a reason; raise, acknowledge and dismiss are all audited.

The test case is a designated vehicle's complete, timestamped route. Cameras whose plates are legible get a continuous reader: a sampled camera does not see a vehicle that crosses in five seconds.

Nobody sees more as cameras are added

1

Brokered, opaque video

Short-lived, watermarked sessions; every segment is re-authorised, so revoking a grant stops a playing stream.

2

Scope in three dimensions

Department, city and zone, checked per request, plus each camera's own access policy.

3

Least privilege

13 roles, 22 permissions. The account that ingests plates cannot read the watchlist back.

4

Withheld, not refused

Without plate permission an officer still gets the alert, with the plate withheld.

5

Retention on read

Plates keep a shorter clock, applied at read time — a failed purge cannot extend it.

6

A complete audit trail

Every sign-in, read, grant, refusal and machine-raised alert, each committed on its own.

Live, measured, and honest about the limits

30/30

grid cameras onboarded from the catalogue, online

36/48

plates a person can read, read correctly, across 38 recorded clips (was 9 of 26)

0

false plates confirmed or sent, on any clip, day or night

332

automated tests passing (201 platform, 131 edge)

Plates read per 2 minutes, grid cam06, by the recording's clock



At noon, 15 of the 16 plates legible by eye were read and none was wrong; the sixteenth is misread by every reader and is refused.

Ground truth is by eye and used only for scoring. At night these 720p feeds carry plates 14–31 px wide: nobody can read them, and the system reads none rather than guessing. A false plate costs more than a missed one — that is why an alert from this system can be acted on.

Built against the grid as it behaves, not as a file

RTSP over TCP

forced in every client; HLS only
when 8554 is blocked

Timing from PTS

never arrival time; the declared
frame rate is never trusted, the
delivered rate is measured

Reconnect with backoff

2 s to 30 s; a failing camera keeps
the full pause, never a tight loop

Decoder warnings on join

logged, not fatal; mixed H.264 and
H.265

The catalogue is the contract

cameras and their properties come
from `/api/ingest`; nothing is hard-
coded

No uniform grid

each camera detected at its own
resolution, with its own profile in
configuration

Scene discontinuity

tracks and motion state reset at the
loop cut; learned traffic flow
survives

Consume only, paced

a bounded pool of streams; nothing
published, no control API

Mapped to the code that does each one in `docs/sentinel-grid.md`.

PLAN FOR SCALE

To ~80,000 cameras without moving the video

Streaming everything centrally

160 Gbps

sustained, and ~1.7 PB of video per day

Vigentra: metadata only

0.6–1 Gbps

statewide, 40–250× less; departments keep their footage

~3,000 GPUs

ANPR only where the optics allow,
classified at onboarding (estimate)

Hot · warm · cold

30 days, 1 year, 7 years; plates on
a shorter clock

RPO 5 min · RTO 1 h

synchronous standby, point-in-
time recovery, DR site

Watched at scale

per-camera health history, liveness
from the video, audited

Named and planned: trigram prefilter for fuzzy plate search, bloom filter for the watchlist, PostGIS for corridor queries.

Each phase is useful on its own

Phase 1 · ~50 cameras

one district, all four models exercised — this hackathon

›

Phase 2 · ~2,000

3 districts, 2 departments — genuinely different VMS vendors

›

Phase 3 · ~15,000

all districts, Home Department — statewide load, DR failover tested

›

Phase 4 · ~80,000

all 26 departments — steady state

Model 1 goes first, in full, at every phase

An accurate statewide camera inventory with coverage-gap analysis is valuable on the day it exists — and nobody has one today. Analytics follow the inventory.

Built, running, and verifiable in the code

Hybrid architecture

Models 1, 3 and 2 together, each where it fits

Cross-camera tracking

routes from plate reads, impossible legs flagged

Analytics beyond ANPR

intrusion zones, wrong-way, stopped in lane, person in traffic

Edge processing

inference beside the video, on CPU or GPU

Bandwidth optimisation

metadata only; video on demand; sampling dial

Security and audit

brokered sessions, fine roles, append-only trail

Automated alerts

matched at ingest, prioritised, signed webhook out

Health monitoring

per-source and per-camera, with history

Operational dashboards

registry, map, live wall, alerts, reports

Integration-ready APIs

documented REST API and adapter contract

What we believe — and what the grid taught us

Start from the people who use it

A control-room officer needs one screen; twenty-six departments should not have to change theirs to give it to them. Federation, not replacement, is what makes that possible on day one.

Never show more certainty than we have

An unconfirmed reading is shown as unconfirmed, a plate too small to read is marked unreadable instead of guessed, and every figure in this deck was measured or is labelled as an estimate.

Privacy is the design, not a feature

Footage is asked for, never ambient. Every look is brokered and audited — refusals included — because the thing that scales worst in surveillance is how many people can see everything.

The hard part was the plumbing

On the government feed the model was rarely the problem: refused sessions, dropped streams and a shared account were. The platform now keeps cameras online while video flows and restarts its own readers.

This is what we were able to build in the time we had.

With upgraded hardware, the same software can do much better.

Today

One laptop — an Apple M1 with 16 GB. Our readers share its single GPU and visit the 30 cameras in turn, a few minutes each.

With edge GPUs

About one T4-class card per 15 cameras (estimate): every camera read continuously, heavier models, more frames of every vehicle.

With cameras placed for ANPR

Plates large enough to confirm rather than guess — on today's wide junction views it is the optics, not the code, that set the ceiling.

Source: github.com/rewardix-ai/vigentra · Design: [docs/hld.md](#) · Scale: [docs/scalability.md](#) · Demonstration videos and output report: links in the submission